RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-01

Single Correct Type Questions:

1. Find the equation of the circle that passes through the points $(1, 0), (-1, 0)$ and $(0, 1)$.

(A) $x^2 + y^2 = 2$ (B) $x^2 + y^2 = 1$

(C) $x^2 - 2x + y^2 = 0$ (D) $x^2 + 2x + y^2 = 2$

MCR166

2. The radius of the circle passing through the points $(1, 2), (5, 2)$ & $(5, -2)$ is

(A) $5\sqrt{2}$ (B) $2\sqrt{5}$

(C) $3\sqrt{2}$ (D) $2\sqrt{2}$

MCR167

3. $ABCD$ is a square in first quadrant whose side is a, taking AB and AD as axes, then the equation to the circle circumscribing the square is

(A) $x^2 + y^2 = a(x + y)$ (B) $x^2 + y^2 = a^2(x + y)$

(C) $x^2 + y^2 = 2a(x + y)$ (D) $x^2 + y^2 = \sqrt{2}a(x + y)$

MCR168

4. The centres of the circles $x^2 + y^2 - 6x - 8y - 7 = 0$ and $x^2 + y^2 - 4x - 10y - 3 = 0$ are the ends of the diameter of the circle

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(A) $x^2 + y^2 - 5x - 9y + 26 = 0$ (B) $x^2 + y^2 + 5x - 9y + 14 = 0$

(C) $x^2 + y^2 + 5x - y - 14 = 0$ (D) $x^2 + y^2 + 5x + y + 14 = 0$

MCR169

5. The circle described on the line joining the points $(0, 1), (a, b)$ as diameter cuts the x-axis in points whose abscissa are roots of the equation:

(A) $x^2 + ax + b = 0$ (B) $x^2 - ax + b = 0$

(C) $x^2 + ax - b = 0$ (D) $x^2 - ax - b = 0$

MCR170

6. A point $A(2, 1)$ is outside the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ & AP, AQ are tangents to the circle. The equation of the circle circumscribing the triangle APQ is :

(A) $(x + g)(x - 2) + (y + f)(y - 1) = 0$

(B) $(x + g)(x - 2) - (y + f)(y - 1) = 0$

(C) $(x - g)(x + 2) + (y - f)(y + 1) = 0$

(D) $(x - g)(x - 2) + (y - f)(y - 1) = 0$

MCR171

7. Find the equation to the circle which passes through the origin and cuts off intercepts equal to 3 & 4 from the positive axes.

(A) $x^2 + y^2 + 3x + 4y = 0$ (B) $x^2 + y^2 - 3x + 4y = 0$

(C) $x^2 + y^2 + 3x - 4y = 0$ (D) $x^2 + y^2 - 3x - 4y = 0$

MCR172

For More Join: @IITJEE_Advanced

8. Find equation of circle which touches x & y axis & perpendicular distance of centre of circle from $3x + 4y + 11 = 0$ is 5. Given that circle lies in Ist quadrant.

(A) $x^2 + y^2 + 4x + 4y + 4 = 0$ (B) $x^2 + y^2 - 4x - 4y + 4 = 0$

(C) $x^2 + y^2 - 4x - 4y + 5 = 0$ (D) $x^2 + y^2 - 4x - 4y + 10 = 0$

MCR173

9. Find the equation to the circle which touches the axis of x at a distance 3 from the origin and intercepts a distance 6 on the axis of y.

(A) $x^2 + y^2 \pm 3\sqrt{2}y \pm 6x + 9 = 0$ (B) $x^2 + y^2 \pm 3\sqrt{2}y \pm 3x + 9 = 0$

(C) $x^2 + y^2 \pm 6\sqrt{2}y \pm 6x + 9 = 0$ (D) $x^2 + y^2 \pm 6\sqrt{2}y \pm 6x + 6 = 0$

MCR174

10. Find equation of circle whose cartesian equation are $x = -3 + 2\sin\theta, y = 4 + 2\cos\theta$

(A) $(x + 3)^2 + (y - 4)^2 = 4$ (B) $(x - 3)^2 + (y - 4)^2 = 5$

(C) $(x + 3)^2 + (y + 4)^2 = 4$ (D) $(x - 3)^2 + (y - 4)^2 = 5$

MCR175

11. The intercepts made by the circle $x^2 + y^2 - 5x - 13y - 14 = 0$ on the x –axis and y –axis are respectively

(A) 9, 13 (B) 5, 13 (C) 9, 15 (D) None

MCR176

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	D	A	A	B	A	D	B	C	A	C

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-02

Single Correct Type Questions:

1. Find the points of intersection of the line $x-y+2=0$ and the circle $3x^2+3y^2-29x-19y+56=0$. Also determine the length of the chord intercepted.

(A) $(1,5),(3,7),4\sqrt{2}$ (B) $(1,3),(5,7),4\sqrt{2}$

(C) $(2,3),(3,7),4\sqrt{2}$ (D) $(1,3),(7,5),4\sqrt{2}$

MCR177

2. Show that the line $7y-x=5$ touches the circle $x^2+y^2-5x+5y=0$ and find the equation of the other parallel tangent.

(A) $x-7y-45=0$ (B) $x-5y-45=0$

(C) $x-7y-35=0$ (D) $x-5y-35=0$

MCR178

3. Find the equation of the normal to the circle $x^2+y^2=2x$, which is parallel to the line $x+2y=3$.

(A) $x-2y-1=0$ (B) $x+2y-1=0$

(C) $x+2y+1=0$ (D) $x-2y+1=0$

MCR179

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4. The equation of normal to the circle $x^2+y^2-4x+4y-17=0$ which passes through $(1,1)$ is

(A) $3x+y-4=0$ (B) $x-y=0$

(C) $x+y=0$ (D) $3x-y-4=0$

MCR180

5. The normal at the point $(3,4)$ on a circle cuts the circle at the point $(-1,-2)$. Then the equation of the circle is

(A) $x^2+y^2+2x-2y-13=0$

(B) $x^2+y^2-2x-2y-11=0$

(C) $x^2+y^2-2x+2y+12=0$

(D) $x^2+y^2-2x-2y+14=0$

MCR181

6. The tangent lines to the circle $x^2+y^2-6x+4y=12$ which are parallel to the line $4x+3y+5=0$ are given by:

(A) $4x+3y-7=0, 4x+3y+15=0$

(B) $4x+3y-31=0, 4x+3y+19=0$

(C) $4x+3y-17=0, 4x+3y+13=0$

(D) $4x+3y-31=0, 4x+3y-19=0$

MCR182

7. The tangent to the circle $x^2+y^2=5$ at the point $(1,-2)$ also touches the circle $x^2+y^2-8x+6y+20=0$ at

(A) $(-2,1)$ (B) $(-3,0)$ (C) $(-1,-1)$ (D) $(3,-1)$

MCR183

For More Join: @IITJEE_Advanced

Multiple Correct Type Questions:

8. The equation of circles passing through $(3, -6)$ touching both the axes is

(A) $x^2 + y^2 - 6x + 6y + 9 = 0$ (B) $x^2 + y^2 + 6x - 6y + 9 = 0$

(C) $x^2 + y^2 + 30x - 30y + 225 = 0$ (D) $x^2 + y^2 - 30x + 30y + 225 = 0$

MCR184

9. Equations of circles which pass through the points $(1, -2)$ and $(3, -4)$ and touch the x-axis is

(A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$

(C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) $x^2 + y^2 + 10x + 20y - 25 = 0$

MCR185

10. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $x - 3y + 1 = 0$. If two vertices of rectangle are the points $(-2, 5)$ and $(6, 5)$ respectively, then which of the following hold(s) good?

(A) Area of rectangle $ABCD$ is 64 square units.

(B) Centre of circle is (2, 1)

(C) The other two vertices of the rectangle are (–2, –3) and (6, – 3)

(D) Equation of sides are $x = -2, y = -3, x = 5$ and $y = 6$.

MCR186

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	A	B	B	D	A,D	B,C	A,B,C

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-03

Single Correct Type Questions:

1. Find the values of p for which the power of a point $(2,5)$ is negative with respect to a circle $x^2+y^2-8x-12y+p=0$ which neither touches the axes nor cuts them.

(A) $P \in (16,47)$ (B) $P \in (16,36)$

(C) $P \in (25,47)$ (D) $P \in (36,47)$

MCR187

2. A line segment through a point P cuts a given circle in 2 points A & B such that $PA=16$ & $PB=9$, find the length of tangent from points to the circle

(A) 7 (B) 25 (C) 12 (D) 8

MCR188

3. If the length of the tangent from (f,g) to the circle $x^2+y^2=6$ be twice the length of the tangent from (f,g) to the circle $x^2+y^2+3x+3y=0$, then $f^2+g^2+4f+4g$ is equal to

(A) –1 (B) –2 (C) 0 (D) None of these

MCR189

4. The length of the tangent drawn from any point on the circle $x^2+y^2+2gx+2fy+p=0$ to the circle $x^2+y^2+2gx+2fy+q=0$ is

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(A) $\sqrt{q-p}$ (B) $\sqrt{p-q}$ (C) $\sqrt{q+p}$ (D) $\sqrt{2q+p}$

MCR190

5. Find the equation of director circle of the circle $(x+4)^2+y^2=8$

(A) $(x-4)^2+y^2=4$ (B) $(x+4)^2+y^2=16$

(C) $(x-4)^2+y^2=16$ (D) $(x+4)^2+y^2=4$

MCR191

6. The locus of the point of intersection of the tangents to the circle $x^2+y^2=a^2$ at points whose parametric angles differ by $\frac{\pi}{3}$ is

(A) $x^2+y^2=\frac{4a^2}{3}$ (B) $x^2+y^2=\frac{2a^2}{3}$

(C) $x^2+y^2=\frac{a^2}{3}$ (D) $x^2+y^2=\frac{a^2}{9}$

MCR192

Multiple Correct Type Questions:

7. The points on the line $x=3$ from which the tangents drawn to the circle $x^2+y^2=7$ are at right angles is/are

(A) $(3,\sqrt{5})$ (B) $(3,-\sqrt{5})$

(C) $(5,\sqrt{3})$ (D) $(5,-\sqrt{3})$

MCR193

For More Join: @IITJEE_Advanced

8. The centre of a circle passing through the points $(0, 0), (1, 0)$ & touching the circle $x^2 + y^2 = 9$ is :

(A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \sqrt{2}\right)$ (C) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, -\sqrt{2}\right)$

MCR194

9. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is $(x - c)^2 + (y - c)^2 = c^2$ where c is

(A) 1 (B) 2 (C) 4 (D) 6

MCR195

10. The equation(s) of the tangent at the point (0, 0) to the circle where circle makes intercepts of length $2a$ and $2b$ units on the coordinate axes, is (are) -

(A) $ax + by = 0$ (B) $ax - by = 0$

(C) $x = y$ (D) $bx + ay = ab$

MCR196

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	B	A	B	A	A,B	B,D	A,D	A,B

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-04

Single Correct Type Questions:

1. Find the equation of the chord of the circle $x^2 + y^2 + 6x + 8y + 9 = 0$ whose middle point is $(-2, -3)$.

(A) $x - y + 3 = 0$ (B) $x + y - 5 = 0$ (C) $x + y + 5 = 0$ (D) $x + y - 3 = 0$

MCR197

2. Tangents are drawn to the circle $x^2 + y^2 = 12$ at the points where it is met by the circle $x^2 + y^2 - 5x + 3y - 2 = 0$; find the point of intersection of these tangents.

(A) $\left(6, -\frac{18}{5}\right)$ (B) $\left(-6, -\frac{18}{5}\right)$ (C) $\left(6, \frac{18}{5}\right)$ (D) $\left(-6, \frac{18}{5}\right)$

MCR198

3. The locus of the centers of the circles such that the point $(2, 3)$ is the mid-point of the chord $5x + 2y = 16$ is

(A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$

(C) $2x + 5y + 11 = 0$ (D) $2x - 5y - 11 = 0$

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MCR199

4. The chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ pass through the point

(A) (1, 2) (B) $\left(\frac{1}{2}, \frac{1}{4}\right)$ (C) (2, 4) (D) (4, 4)

MCR200

5. The locus of the mid-point of a chord of the circle $x^2 + y^2 = 4$ which subtends a right angle at the origin is

(A) $x + y = 2$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = 2$ (D) $x + y = 1$

MCR201

6. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

MCR202

7. Circles $C_1: x^2 + y^2 = 9$ & $C_2: x^2 + y^2 - 4x = 0$. Common chord PQ of C_1 & C_2 intersects the direct common tangent at R. If $RP + RQ = \sqrt{\frac{a}{b}}$, where a & b are relatively prime. Then $(a + b)$ equals

(A) 65 (B) 63 (C) 71 (D) 79

MCR203

For More Join: @IITJEE_Advanced

Multiple Correct Type Questions:

8. For circle $x^2 + y^2 = 16$. Locus of a point P such that area enclosed by tangents drawn from P to circle and chord of contact is 8 is curve C_1. Then area of curve C_1 is $16k\pi$ where -

(A) k is even integer (B) k is a prime number

(C) k is irrational (D) k is odd integer

MCR204

9. Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0, 7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio $2 : 1$.

(A) $3x - 4y - 25 = 0$ (B) $4x + 3y - 25 = 0$

(C) $4x - 3y - 25 = 0$ (D) $3x + 4y - 25 = 0$

MCR205

Matrix Match Type Questions:

10.

	Column – I	**Column – II**
(A)	Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is	(p) 0
(B)	The number of circles touching all the three lines $3x + 7y = 2, 21x + 49y = 5$ and $9x + 21y = 0$ are	(q) 2
(C)	The length of common chord of circles $x^2 + y^2 - x - 11y + 18 = 0$ and $x^2 + y^2 - 9x - 5y + 14 = 0$ is	(r) 5
(D)	Number of common tangents of the circles $x^2 + y^2 - 2x = 0$ and $x^2 + y^2 + 6x - 6y + 2 = 0$ is	(s) 3

MCR206

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	A	B	C	C	D	A,B	C,D	A→ q; B → p; C → r: D →s

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-05

Single Correct Type Questions:

1. If $y = 2x$ is a chord of the circle $x^2 + y^2 - 10x = 0$, find the equation of a circle with this chord as diameter.

(A) $x^2 + y^2 - x - 2y = 0$ (B) $x^2 + y^2 - 2x - 4y = 0$

(C) $x^2 + y^2 - x + 2y = 0$ (D) $x^2 + y^2 + 2x - y = 0$

MCR207

2. Find the equation of a circle which touches the line $2x - y = 4$ at the point $(1, -2)$ and passes through $(3, 4)$

(A) $(x-1)^2 + (y+2)^2 + 10\,(2x - y - 4) = 0$

(B) $(x-1)^2 + (y+2)^2 + 15\,(2x - y - 4) = 0$

(C) $(x-1)^2 + (y+2)^2 + 20\,(2x - y - 4) = 0$

(D) $(x-1)^2 + (y+2)^2 + 25\,(2x - y - 4) = 0$

MCR208

3. Find the equation of a circle which touches the line $2x - y = 4$ at the point $(1, -2)$ and radius $= 5$

(A) $(x-1)^2 + (y+2)^2 \pm \sqrt{2}(2x - y - 4) = 0$

(B) $(x-1)^2 + (y+2)^2 \pm \sqrt{10}\,(2x - y - 4) = 0$

(C) $(x-1)^2 + (y+2)^2 \pm \sqrt{15}\,(2x - y - 4) = 0$

(D) $(x-1)^2 + (y+2)^2 \pm \sqrt{20}\,(2x - y - 4) = 0$

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MCR209

4. Show that the equation $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ represents for different values of λ a system of circles passing through two fixed points A and B on the x –axis, and also find the equation of that circle of the system the tangent to which at A and B meet on the line $x + 2y + 5 = 0$.

(A) $x^2 + y^2 - 2x - 6y - 8 = 0$ (B) $x^2 + y^2 - x - 3y - 8 = 0$

(C) $x^2 + y^2 + 2x + 6y - 8 = 0$ (D) $x^2 + y^2 + x + 3y - 8 = 0$

MCR210

5. Find the equation of the circle which passes through the point $(1, 1)$ & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point $(2, 3)$ on it.

(A) $x^2 + y^2 + x - 6y + 3 = 0$ (B) $x^2 + y^2 + x - 6y - 3 = 0$

(C) $x^2 + y^2 + x - 6y + 3 = 0$ (D) $x^2 + y^2 + x - 3y + 3 = 0$

MCR211

6. Equation of the circle which passes through the point $(-1, 2)$ & touches the circle $x^2 + y^2 - 8x + 6y = 0$ at origin, is -

(A) $x^2 + y^2 - 2x - \frac{3}{2}y = 0$ (B) $x^2 + y^2 + x - 2y = 0$

(C) $x^2 + y^2 + 2x + \frac{3}{2}y = 0$ (D) $x^2 + y^2 + 2x - \frac{3}{2}y = 0$

MCR212

For More Join: @IITJEE_Advanced

7. Find the equations of smallest circle which passes through the points of intersection of the line $x + y = 1$ and the circle $x^2 + y^2 = 9$.

(A) $x^2 + y^2 - 2x - 2y - 8 = 0$

(B) $x^2 + y^2 + 2x - 2y - 8 = 0$

(C) $x^2 + y^2 - x + y - 8 = 0$

(D) $x^2 + y^2 - x - y - 8 = 0$

MCR213

Multiple Correct Type Questions:

8. If $(a, 0)$ is a point on a diameter segment of the circle $x^2 + y^2 = 4$, then $x^2 - 4x - a^2 = 0$ has

(A) exactly one real root in $(-1, 0]$

(B) Exactly one real root in $[2, 5]$

(C) distinct roots greater than -1

(D) Distinct roots less than 5

MCR214

Matrix Match Type Questions:

9.

	Column – I	Column – II
(A)	If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to	(p) 13
(B)	Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of hormonic mean between squares of their radii, then k^4 equals to	(q) 7
(C)	The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda^2}{4}$, then the value of λ is	(r) 4
(D)	The number of integral points which lie on or inside the circle $x^2 + y^2 = 4$ is	(s) 2

MCR215

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	B	C	D	A	A	D	D	A,B,C,D	A→ s, B → r, C → q, D →p

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-06

Single Correct Type Questions:

1. Find the equations of the transverse common tangents of the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ & $x^2 + y^2 + 6x - 2y + 1 = 0$

(A) $y = 0, 4x + 3y = 10$ (B) $x = 0, 4x - 3y = 10$

(C) $x = 0, 3x + 4y = 10$ (D) $y = 0, 3x + 4y = 10$

MCR216

2. Find the equations to the direct common tangents of the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ & $x^2 + y^2 + 6x - 2y + 1 = 0$

(A) $y = 4$ and $3y = 4x$ (B) $y = 4$ and $4y = 3x$

(C) $x = 4$ and $3y = 4x$ (D) $x = 4$ and $4y = 3x$

MCR217

3. Number of common tangents of the circles $(x + 2)^2 + (y - 2)^2 = 49$ and $(x - 2)^2 + (y + 1)^2 = 4$ is

(A) 0 (B) 1

(C) 2 (D) 3

MCR218

4. If the length of a common internal tangent to two circles is 7, and that of a common external tangent is 11, then the product of the radii of the two circles is

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(A) 18 (B) 20

(C) 16 (D) 12

MCR219

5. The length of transverse common tangent of the circles $x^2 + y^2 = 1$ and $(x - h)^2 + y^2 = 1$ is $2\sqrt{3}$, then the value of h is :

(A) ± 2 (B) ± 4

(C) $\pm\sqrt{3}$ (D) ± 6

MCR220

6. If d denotes the distance of origin from the common tangent to circles S_1 & S_2 where $S_1 : x^2 + y^2 - 2x - 4y = 44$ & $S_2 : x^2 + y^2 - 8x - 12y + 48 = 0$, then $[d]$ is equal to

(where $[\cdot]$ denotes greatest integer function)

(A) 3 (B) 6

(C) 9 (D) 10

MCR221

7. Circle K is circle of largest radius, inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K, is -

(A) $\frac{6 - \sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$

(C) 3 (D) $6\left(\sqrt{2} - 1\right)$

MCR222

For More Join: @IITJEE_Advanced

Multiple Correct Type Questions:

8. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.

(A) Number of integral values of r are 14 for which circles are intersecting.

(B) Number of integral values of r are 9 for which circles are intersecting.

(C) For r equal to 13 number of common tangents are 3.

(D) For r equal to 21 number of common tangents are 2.

MCR223

9. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?

(A) S_1 and S_2 intersect at an angle of 90°.

(B) The point of intersection of the two circle are (2, 0) and $\left(\frac{6}{5}, \frac{8}{5}\right)$.

(C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.

(D) The point (2, 3) lies outside the circles S_1 and S_2.

MCR224

Subjective Type Questions:

10. A circle $S = 0$ is drawn with its centre at (–1, 1) so as to couch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle S = 0 on the coordinate axies.

MCR225

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	C	A	B	A	B	C	D	A,C	A,C,D	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Circle: RACE-07

Single Correct Type Questions:

1. The circles $x^2 + y^2 - 2x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 6 = 0$ cut each other at
(A) 60° (B) 30° (C) 45° (D) 90°

MCR226

2. Find the equation of the circle passing through the origin and cutting the circles $x^2 + y^2 - 4x + 6y + 10 = 0$ and $x^2 + y^2 + 12y + 6 = 0$ at right angles.
(A) $(x^2 + y^2) - 7x + 2y = 0$ (B) $2(x^2 + y^2) - 7x + 2y = 0$
(C) $(x^2 + y^2) - 14x + 4y = 0$ (D) $2(x^2 + y^2) + 7x - 2y = 0$

MCR227

3. Given the three circles $x^2 + y^2 - 16x + 60 = 0, 3x^2 + 3y^2 - 36x + 81 = 0$ and $x^2 + y^2 - 16x - 12y + 84 = 0$,
find the point from which the tangents to them are equal in length and find this length.
(A) $\left(\frac{33}{4}, 2\right)$; $L_T = \frac{1}{4}$ (B) $\left(\frac{33}{2}, 2\right)$; $L_T = \frac{1}{2}$
(C) $\left(\frac{33}{4}, 4\right)$; $L_T = \frac{1}{4}$ (D) $\left(\frac{11}{4}, 2\right)$; $L_T = \frac{1}{2}$

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4. Equation of the circle cutting orthogonally the three circles $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$ is
(A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$
(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) $x^2 + y^2 + 16x - 18y - 4 = 0$

MCR229

5. The line $x + 2y + 3 = 0$ cuts the circle $x^2 + y^2 + 4x + 4y - 1 = 0$ at points P and Q and the line $2x + 3y + \lambda = 0$ cuts the circle $x^2 + y^2 + 6x + 2y - 7 = 0$ at points R and S. If P, Q, R, S are concyclic, then the value of λ is
(A) 2 (B) 4 (C) 6 (D) 8

MCR230

Comprehension Type Questions:

Comprehension # 1 (Q. No. 6 to 8)

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0, x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, then

6. Point from which length of tangents to these three circles is same is
(A) (1, 0) (B) (3, 2)
(C) (10, 5) (D) (– 2, 1)

MCR231

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7. Equation of circle S_4 which cut orthogonally to all given circle is

(A) $x^2 + y^2 - 6x + 4y - 14 = 0$ (B) $x^2 + y^2 + 6x + 4y - 14 = 0$

(C) $x^2 + y^2 - 6x - 4y + 14 = 0$ (D) $x^2 + y^2 - 6x - 4y - 14 = 0$

MCR232

8. Radical centre of circles S_1 , S_2, & S_4 is

(A) $\left(-\frac{3}{5}, -\frac{8}{5}\right)$ (B) (3, 2) (C) (1, 0) (D) $\left(-\frac{4}{5}, -\frac{3}{2}\right)$

MCR233

Comprehension # 2 (Q. No. 9 to 11)

Two circles are

$S_1 \equiv (x+3)^2 + y^2 = 9$

$S_2 \equiv (x-5)^2 + y^2 = 16$

with centres C_1 & C_2

9. A direct common tangent is drawn from a point P (on x-axis) which touches S_1 & S_2 at Q & R, respectively. Find the ratio of area of ΔPQC_1 & ΔPRC_2 .

(A) 3 : 4 (B) 9 : 16 (C) 16 : 9 (D) 4 : 3

MCR234

10. From point $'A'$ on S_2 which is nearest to C_1 , a variable chord is drawn to S_1. The locus of mid point of the chord.

(A) circle (B) Diameter of s_1

(C) Arc of a circle (D) chord of s_1 but not diameter

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11. Locus obtained in question above cuts the circle S_1 at B & C, then line segment BC subtends an angle on the major arc of circle S_1 is

(A) $\cot^{-1}\frac{3}{4}$ (B) $\sin^{-1}\frac{3}{4}$

(C) $\tan^{-1}\frac{3}{4}$ (D) $\cot^{-1}\frac{3}{4}$

MCR236

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	B	A	A	B	B	D	A	B	C	A

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